CHAPTER 4

Chemical Bonding and Molecular Structure

Sr. No.	Concept	Formulas	Other information
1.	**Formal charge**	Number of valence shell electrons – Number of bonds – Number of unshared electrons	N.A.
2.	**Hydration energy**	Hydration energy $\propto q_+q_-$ Hydration energy $\propto \frac{1}{r_+}+\frac{1}{r_-}$	Here, q_+ and q_- are charge of cation and anion respectively. r_+ and r_- are ionic radius of cation and anion respectively.
3.	**Lattice energy**	Lattice energy $\propto q_+q_-$ $\propto \frac{1}{r_+ + r_-}$	Here, q_+ and q_- are charge of cation and anion respectively. r_+ and r_- are ionic radius of cation and anion respectively.
4.	**Solubility**	Solubility $\propto$ Dielectric constant Solubility $\propto \frac{1}{\text{Lattice energy}}$ Solubility $\propto$ Hydration energy	N.A.
5.	**Fajan's rule** (a) Polarization	Covalent character $\propto$ Polarization $\phi = \frac{\text{Charge on cation}}{\text{Size of cation}}$ Charge on cation or anion $\propto$ polarization $\propto$ covalent character Size of cation $\propto \frac{1}{\text{Polarization}} \propto \frac{1}{\text{Covalent character}}$ Size of anion $\propto$ polarization $\propto$ covalent character	Here, Z_{eff} is effective nuclear charge. ϕ is ionic character.
6.	**Valence bond theory**	Extent of overlapping p – p > s – p > s – s Overlapping $\propto \frac{1}{\text{size of orbital}}$	Here, s and p are orbitals.

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7.	**Valence shell electron pair repulsion theory (VSEPR)**	Order of repulsion: L.P – L.P. > L.P. – B.P. > B.P. – B.P.	Here, L.P. is lone pair. B.P. is bond pair.
8.	**Hybridisation** (a) Hybridisation number	Hybridisation number = Number of σ bonds + lone pair	N.A.
9.	**Dipole moment (μ)**	$\mu = q \times d$	Here, q is charge d is distance of separation. 1 Debye = 10^{-18} esu-cm. 1D = 3.33564×10^{-30}cm
10.	**Hydrogen bonding**	Strength of H-bonding $\propto$ E.N. of highly electronegative element Strength of intermolecular H-bond > Intramolecular bond.	Here, E.N. is electronegativity.
11.	**Molecular Orbital Theory**	Energy of ABMO > Energy of A.O. > energy of BMO	Here, ABMO is anti-bonding molecular orbital BMO is bonding molecular orbital A.O. is atomic orbital
11.	**Bonding parameters** (a) Bond order (B.O.) (b) Bond length (B.L.) (c) Bond angle (B.A.)	Bond order $=\frac{1}{2}[N_b - N_a]$ Stability $\propto$ B.O. $\propto$ bond dissociation energy $\text{B.O.} \propto \frac{1}{\text{Bond Length (B.L.)}}$ B.L. $\propto$ Atomic size $\text{B.L.} \propto \frac{1}{\text{s} - \text{character}}$ B.A. $\propto$ % s-character $\text{B.A.} \propto \frac{1}{\text{Number of lone pair}}$ (when hybridization is same) Bond angle $\propto$ E.N. of central atom (when hybridization is same) $\text{B.A.} \propto \frac{1}{\text{E.N. of side atom}}$ (when hybridization is same)	Here, N_a = number of antibonding electrons N_b = number of bonding electrons If $N_b > N_a$ B.O. = +ve molecule exist $N_b < N_a$ B.O. = –ve molecule does not exist. E.N. is electronegativity.